

Analgesic Activity of the Methanolic Extract and Total Alkaloids of *Glaucium paucilobum*

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SUMMARY

The species of *Glaucium* have been used in Iranian herbal medicine as laxative, hypnotic, narcotic, and antidiabetic agents and also in the treatment of dermatitis. The analgesic activity of the aerial parts of *Glaucium paucilobum* Freyn (Papaveraceae), a native plant of Iran, were studied using formalin, hot plate, and writhing tests in rodents. The methanolic extract and total alkaloids of *Glaucium paucilobum* caused graded inhibition of both phases of formalin-induced pain. In the hot plate test, i.p. administration of *G. paucilobum* extract at the doses of 50–90 mg/kg and total alkaloids at the dose of 10–60 mg/kg significantly raised the pain threshold at an observation time of 30 min in comparison with control ($p < 0.001$). In the writhing test, the extract at the doses of 30–90 mg/kg and total alkaloids at the doses of 10–60 mg/kg produced a significant decrease in the number of writhings in comparison with the control group ($p < 0.001$). The methanolic extract and total alkaloids of *G. paucilobum*, at antinociceptive doses, did not affect the motor coordination of animals when assessed in the rotarod model. The results showed that analgesic activity of this plant may be related partly to the alkaloids. © 2006 Prous Science. All rights reserved.

Key words: Alkaloids - Formalin test - *Glaucium paucilobum* - Hot plate test - Rotarod test - Writhing test

INTRODUCTION

The genus *Glaucium* (Papaveraceae) comprises 11 species, which are found wild in some regions of Iran (1). The species of *Glaucium* has been used in Iranian herbal medicine as laxative, hypnotic, narcotic, and antidiabetic agents. The leaves of the species of *Glaucium* are also used in the treatment of dermatitis (2). We have already reported several alkaloids from *Glaucium* genus (3-7). In 2002, the antiinflammatory and analgesic activities and acute toxicity of the methanolic extract of *G. grandiflorum* were evaluated (8). In 2003, the antifungal activity of the methanolic extract and alkaloids of *G. oxylobum* were also investigated (9). In 2004, the anti-inflammatory and analgesic activities of the topical preparation of *G. grandiflorum* were also studied (10). In 2005, the antibacterial activity of *G. grandiflorum*, *G. oxylobum* and *G. paucilobum* was reported (11). *G. paucilobum* Freyn is a biennial plant scattered in the northeast of Iran. The plant blooms from May until the end of July and reaches the height of 40–60 cm (12). A bibliographical survey showed that there was no report on the analgesic activity of *G. paucilobum*. Since some natural alkaloids show analgesic activity (13, 14), we decided to investigate the analgesic activity of the methanolic extract and total alkaloids of *G. paucilobum*, as a continuation of the studies of Iranian wild species of the Papaveraceae family.

The present paper describes some of the pharmacological activities of this plant, using formalin, hot plate, writhing, and rotarod tests in rodents.

MATERIALS AND METHODS

Plant material

Glaucium paucilobum Freyn was collected from the suburb of Bujnurd, northeast of Iran, in May 2003 and identified by Dr. Gh. Amin, Department of Pharmacognosy, Tehran University of Medical Sciences. A voucher specimen (No. 6505) was preserved at the herbarium of Faculty of Pharmacy, Tehran University of Medical Sciences. The aerial parts of plant were dried in the shade and powdered so that all the material could be passed through a mesh not larger than 0.5 mm.

Extraction procedures

Powdered plant material (800 g) was soaked in 7 l of methanol for 1 day, and the step was repeated twice, followed by filtration through filter paper. The filtrate was evaporated to dryness under reduced pressure and weighed (71.20 g, 8.90%). The dried extract (60 g) was fractionated between 50% aqueous acetic acid and petroleum ether. The aqueous acidic layer extracted with CHCl_3 . The residue was evaporated to dryness under reduced pressure to give 5.8 g of a crude mixture of alkaloids.